Homework 3

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Part 1. Set up tasks

```
# reading in packages
library(tidyverse)
library(janitor)
library(here)

# reading in kelp data
kelp <- read.csv(
  here("data", "temp-kelp.csv")
)

# reading in personal data finalized
personal_data <- read.csv(
  here("data", "personal_data.csv")
)
```

Part 2. Problems

Problem 1. Giant kelp fronds

Climate change threatens all ecosystems, including giant kelp (*Macrocystis pyrifera*) forests.

You are interested in the potential relationship between ocean temperature (measured in °C) and giant kelp frond elongation rate (also known as growth rate, measured in cm day^{-1}).

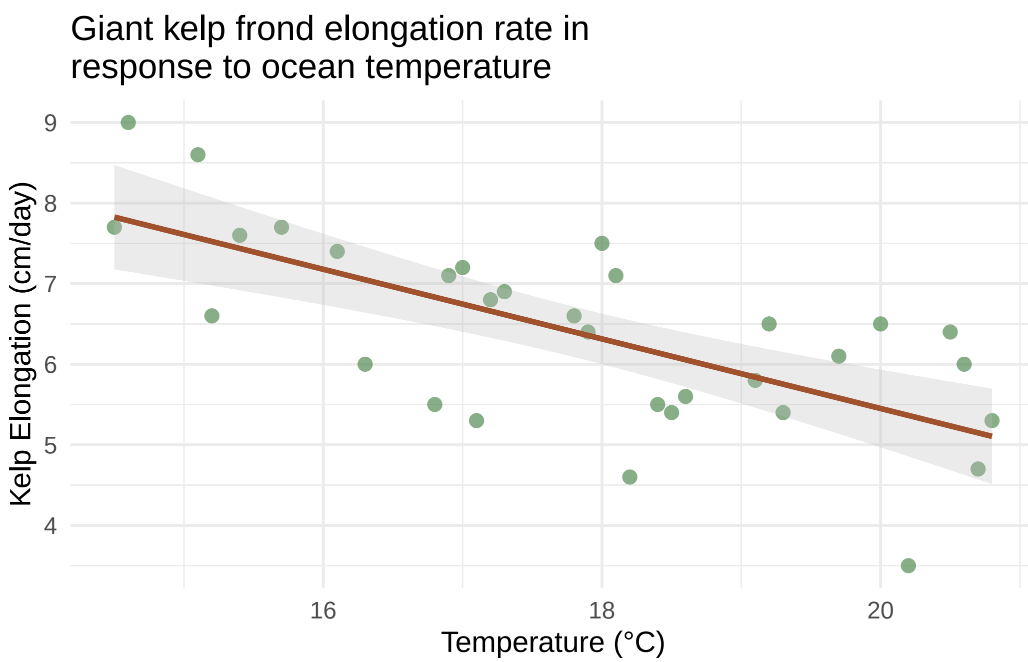
You conduct a field study in which you track giant kelp individuals growing in different temperatures (on average) and measure their elongation rates. Admittedly, this is not a perfect study, but you do the best with what you have!

a. An appropriate test

The appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are Pearson's correlation (r) and Spearman's rank correlation (ρ) test. Pearson's correlation (r) evaluates a linear relationship between the continuous variables and assumes normality of the data, whereas Spearman's rank correlation (ρ) assesses the monotonic relationship by ranking the data points. Since biological growth rates like kelp elongation often do not follow a typical linear pattern, Spearman's rank correlation (ρ) may provide a more accurate measure of the relationship strength.

b. Create a visualization

```
# base layer: ggplot
ggplot(kelp, # from data source
       mapping = aes(
         x = temp_c, # x axis temp
         y = kelp_elong # y axis elongation
       )) +
  # first layer : data points showing individual observation
  geom_point(size = 2,
             alpha = 0.7,
             color = "palegreen4") + # adding custom color
  # adding a linear trend line
  # adding ribbon to represent confidence interval
  geom_smooth(method = "lm",
              color = "sienna",
              fill = "gray",
              alpha = 0.3) +
  # renaming axes, adding a title
  labs(
    x = "Temperature (°C)",
    y = "Kelp Elongation (cm/day)",
    title = "Giant kelp frond elongation rate in
response to ocean temperature"
  ) +
  # adding theme
  theme_minimal()
```



Ocean temperature increase leads to an approximate decrease in kelp elongation.

c. Check your assumptions and run your test

Checking assumptions for Pearson's correlation (r)

```
# writing out model in code
kelp_model <- lm(
  kelp_elong ~ temp_c,
  data = kelp)
# looking at model results
summary(kelp_model)
```

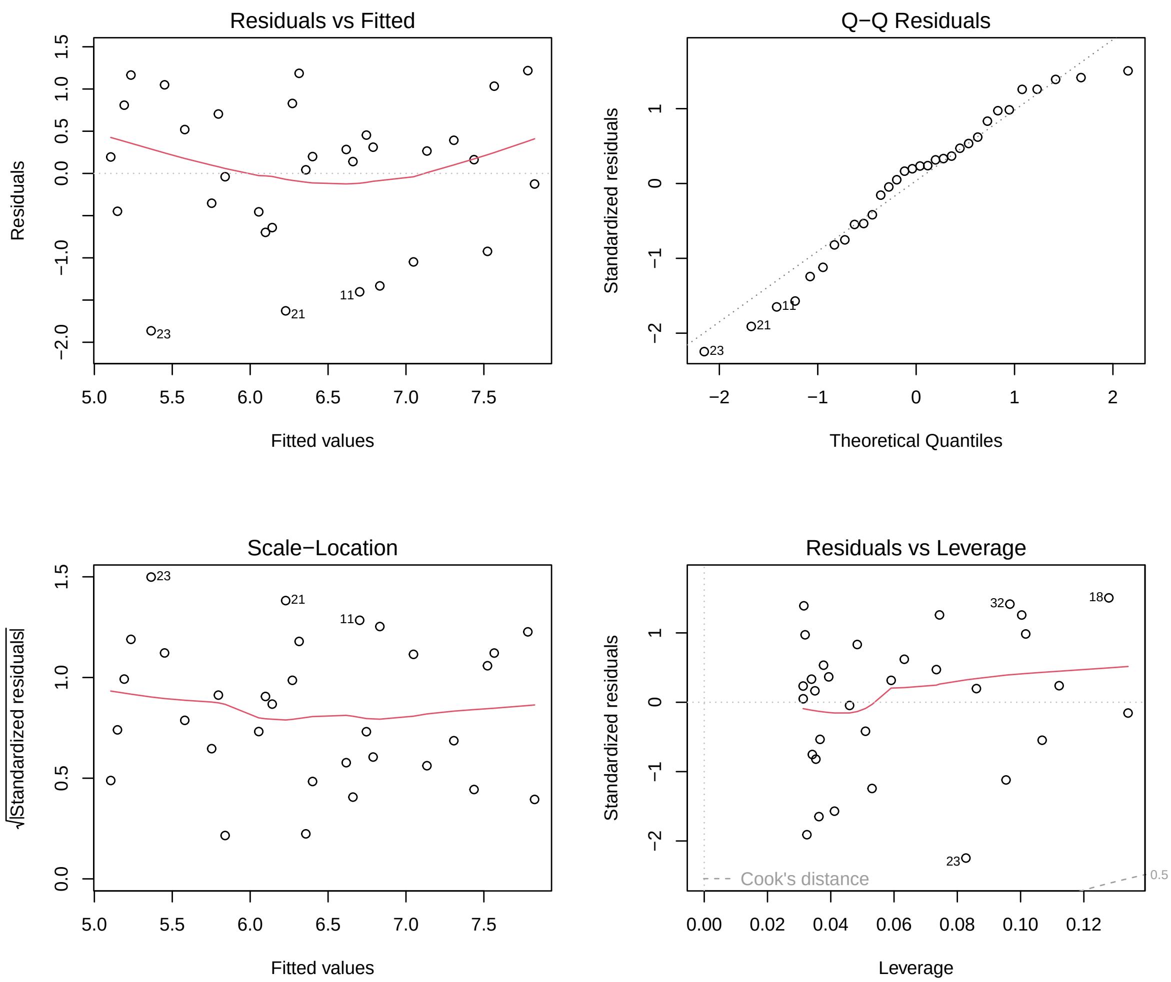
```
Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max
-1.8643 -0.5017  0.1790  0.5659  1.2177
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473, Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

```r
# looking at diagnostics
par(mfrow = c(2,2)) # creating 2 x 2 grid
plot(kelp_model) # plotting diagnostics
```



Diagnostic Assessment

- **Residuals vs Fitted & Scale-Location:** The residuals vs fitted and scale-location appear to be homoscedastic through the approximate match of the red line to the trend line and scatter of points in relation to the line, indicating constant variance.
- **QQ-Residiuals:** The QQ residuals points approximately follow the reference line with slight deviation at the left and right tails, allowing the assumption of normality to be met.
- **Residuals vs Leverage:** No outliers are past Cook's distance, indicating that outliers are not influencing the model coefficient estimates.

Since the assumptions of normality and homoscedasticity are met, and no influential outliers are present, a linear model framework is appropriate for this data, validating the use of a

Pearson's correlation test (r).

Running a Pearson's correlation test (r)

```r
# creating a new object to display correlation test results for pearson's correlation test
kelp_pearson <- cor.test(kelp$temp_c,
                         kelp$kelp_elong,
                         # pearson's correlation
                         method = "pearson")
# viewing correlation test results
kelp_pearson
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between between temperature and giant kelp frond elongation rate I used a Pearson's correlation test (r) due to the model matching assumptions of linearity through approximate normality and homoscedasticity and no outliers being determined to influence the model coefficients. We found a moderate-strong relationship between ocean temperature and kelp elongation (Pearson's r = -0.69, 95% CI $[-0.84, -0.45]$, t(30) = -5.19, p $<$ 0.001, alpha = 0.05). This indicates that there is a negative relationship between temperature and kelp elongation where for every 1 $^{\circ}$C increase, kelp elongation decreases by 0.43 cm day^{-1}.

e. Test implications

Our analysis confirms a significant, moderate-strong negative relationship between ocean warming and giant kelp productivity, with kelp frond elongation rates decrease by 0.43 cm day^{-1} for every 1 $^{\circ}$C increase. This decline indicates that increasing ocean temperatures are directly damaging giant kelp habitats, which without proper action may result in greater loss of biodiversity as species that rely on giant kelp forests become at risk. To protect this ecosystem, a prioritization of restoration and monitoring of giant kelp habitats experiencing high thermal stress.

f. Double check your own work

Running a Spearman's rank correlation test

```
# creating a new object to display correlation test results for spearman's rank test
kelp_spearman <- cor.test(kelp$temp_c,
                          kelp$kelp_elong,
                          # spearman's correlation
                          method = "spearman")
# viewing correlation test results
kelp_spearman
```

```

    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

For both tests I found a moderate-strong relationship between ocean temperature and kelp elongation through the ρ and r value of -0.69, and a rejection of the null hypothesis with a p $<$ 0.001. Indicating that whether the data is analyzed through the monotonic relationship or linearly, we get the same conclusion that increasing ocean temperatures correlate with a decline in in kelp growth.

Problem 2. Personal data

a. Updating visualizations

Figure A: Productivity Score By Location

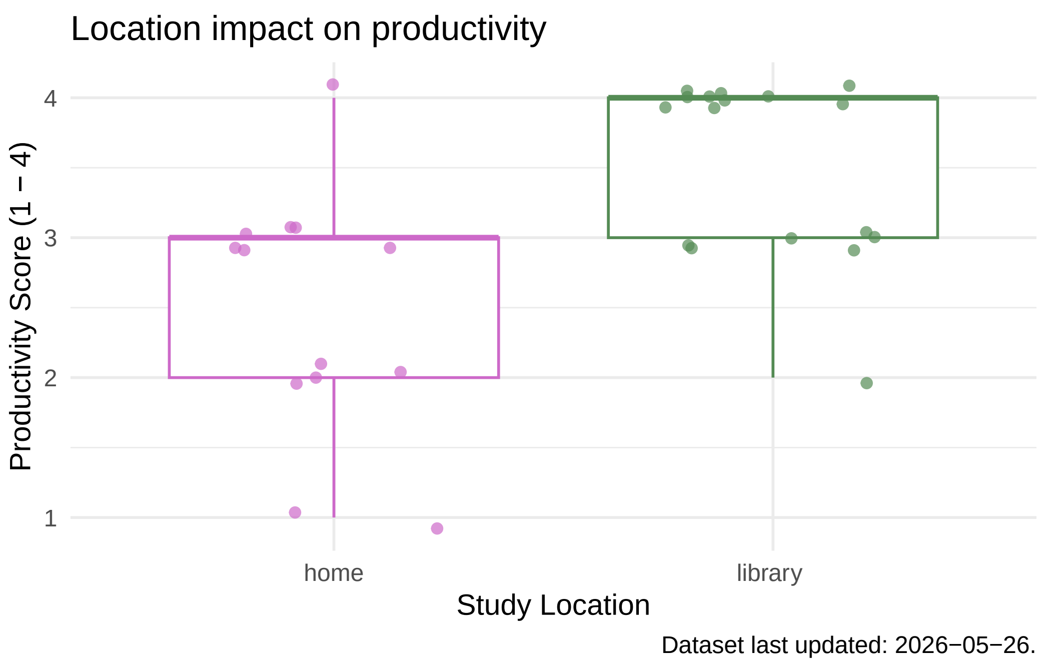
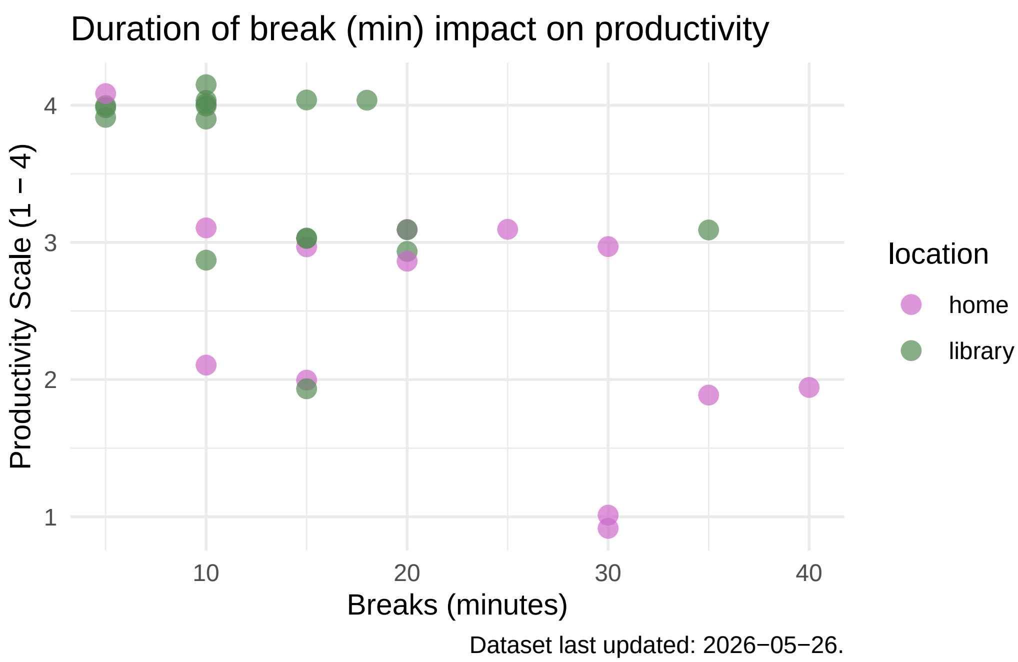


Figure B: Productivity Score By Breaks



b. Captions

Figure A: Productivity scores differ across study environments. Jittered points represent individual observations from self reported study sessions (orchid: home, green: library), with data current through May 26, 2026. Productivity is evaluated on a tiered percentage scale where 1 indicates 50% or less productivity, 2 indicates 50–67%, 3 indicates 67–83%, and 4 indicates 83% or more. Overlaid box plots display a summary of the data, where horizontal bolded lines indicate median productivity score for each location, while the upper and lower box boundaries represent the interquartile range (IQR).

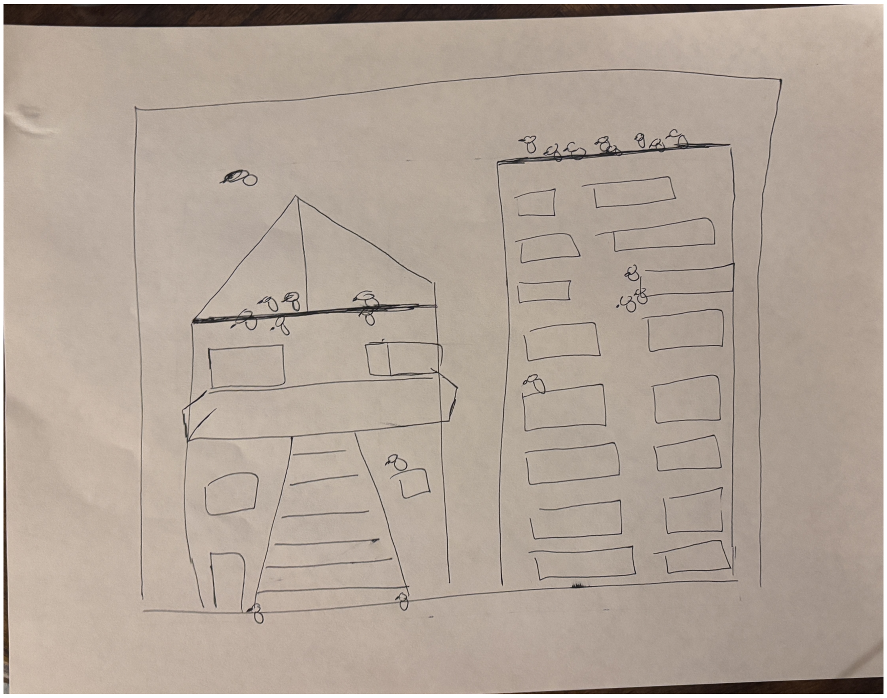
Figure B: Productivity scores differ across varying break durations and study environments Individual points represent distinct recorded study sessions, with data current through May 26, 2026. Productivity is evaluated on a tiered percentage scale where 1 indicates 50% or less productivity, 2 indicates 50–67%, 3 indicates 67–83%, and 4 indicates 83% or more. Point colors correspond to the session location (orchid: home, pale green: library), plotted continuously along the x-axis to show how the duration of breaks taken (in minutes) relates to subsequent productivity ratings (1–4).

Problem 3. Affective Visualization

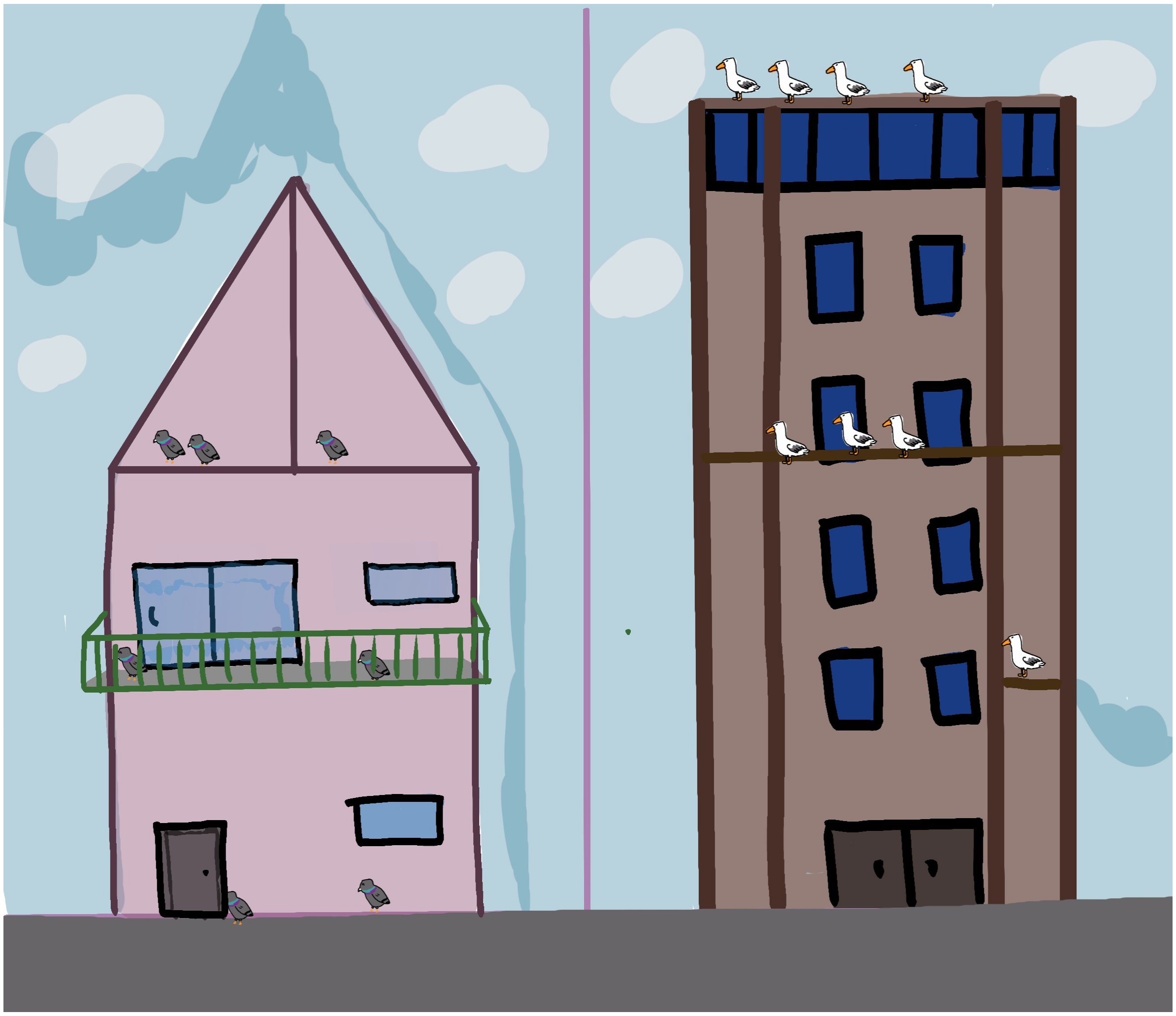
a. Describe in words what an affective visualization could look like for your personal data (3-5 sentences).

An affective visualization for my personal data may look like drawings of my comparison locations within my visualization. Through utilizing the outline of my boxplots generated in Figure A, I can use the IQR line of the home data set as the roof peak, and the median line for both data sets as details of the buildings. For the home drawing I will include details of my current home to indicate the personal nature of this data. For the library drawing I will attempt to make it look similar to the outside of the UCSB Davidson library to again depict how this data correlates to my life. Individual data points will be indicated by pigeons (home) and seagulls (library).

b. Create a sketch (on paper) of your idea.



c. Make a draft of your visualization.



d. Write an artist statement.

e. Prep your materials to share in class.